The Osacra-Gravaser Framework: Nonlinear Backreaction, Fractional Horizons, and Topological Matter

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Abstract

We propose a novel, non-perturbative formulation of quantum gravity where the linear propagation of a gravitational wave and its trace-free quadratic polarization weight are inextricably bound via the dimensionally-balanced fractional coherent equation $\ell_p^{1/2}\Box^{3/4}h^{(1)}_{\mu\nu} + R^{(2)}_{\mu\nu} = 0$. By structuring the macroscopic event horizon as a discrete fractal manifold (Hausdorff dimension $D_H = 3/2$), we map the system to fractional calculus. This approach resolves ultraviolet divergences by bottlenecking momentum-space degrees of freedom to scale as $\Lambda^{3/2}$, while strictly preserving the holographic principle ($S \propto A$). Time emerges as the discrete accumulation of these entangled polarization layers via a thermodynamic bridge, and matter manifests as topological knots piercing this stacked entanglement driven by a continuous trace-free polarization gradient.

1 Introduction

In standard perturbative quantum gravity, the metric tensor $g_{\mu\nu}$ is expanded around a flat Minkowski background $\eta_{\mu\nu}$ as $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. Historically, the orders of this perturbation are solved sequentially, treating the resulting interactions as higher-order loop corrections, which leads to non-renormalizable infinities. The Osacra-Gravaser Framework explicitly binds the first and second orders into a continuous nonlinear backreaction, defining each point in spacetime not as a continuous coordinate, but as a discrete entropic pairing structured upon a non-integer manifold.

2 The Fractional Coherent Equation and Backreaction

We begin by deriving the exact tensor expansion that justifies the fundamental binding. Using the inverse metric expansion $g^{\mu\nu} \approx \eta^{\mu\nu} - h^{\mu\nu} + h^{\mu}_{\alpha}h^{\alpha\nu}$, we expand the Christoffel symbols $\Gamma^{\alpha}_{\mu\nu}$ to second order:

$$\Gamma^{(1)\alpha}_{\mu\nu} = \frac{1}{2}\eta^{\alpha\beta}(\partial_\mu h_{\nu\beta} + \partial_\nu h_{\mu\beta} - \partial_\beta h_{\mu\nu}) \tag{1}$$

The Ricci tensor is defined as $R_{\mu\nu} = \partial_\alpha \Gamma^{\alpha}_{\mu\nu} - \partial_\nu \Gamma^{\alpha}_{\mu\alpha} + \Gamma^{\alpha}_{\alpha\beta}\Gamma^{\beta}_{\mu\nu} - \Gamma^{\alpha}_{\nu\beta}\Gamma^{\beta}_{\mu\alpha}$. At first order, applying the harmonic de Donder gauge ($\partial_\mu h^{\mu}_{\nu} = \frac{1}{2}\partial_\nu h^{\alpha}_{\alpha}$), the linear curvature simplifies to the wave operator:

$$R^{(1)}_{\mu\nu} = -\frac{1}{2}\Box h^{(1)}_{\mu\nu} \tag{2}$$

By expanding to second order, we extract the trace-free quadratic polarization weight, $R^{(2)}_{\mu\nu}$, which represents the self-gravitating energy of the wave:

$$R^{(2)}_{\mu\nu} = \frac{1}{2}h^{\alpha\beta}(\partial_\mu\partial_\nu h_{\alpha\beta} + \partial_\alpha\partial_\beta h_{\mu\nu} - \partial_\alpha\partial_\nu h_{\mu\beta} - \partial_\alpha\partial_\mu h_{\nu\beta}) + \mathcal{O}(\partial h \partial h) \tag{3}$$

Because our manifold operates on a non-integer fractal geometry, the standard d'Alembertian must be upgraded to a Fractional d'Alembertian ($\Box^{3/4}$) to account for non-local information transfer across fractal gaps. To resolve the resulting dimensional mismatch, we introduce a fundamental length scale parameter ℓ_p (analogous to the Planck length). In a vacuum, setting $R_{\mu\nu} = 0$ yields our localized coherent equation:

$$\ell_p^{1/2} \Box_{(n)}^{3/4} h_{\mu\nu}^{(1,n)} + R_{\mu\nu}^{(2,n)} = 0 \tag{4}$$

This establishes that the linear wave ($h^{(1)}$) is shaped by its own gravitational weight ($R^{(2)}$) within a discrete fractal layer n.

3 Fractal Horizons and Taming UV Divergences

To regulate the infinite degrees of freedom characteristic of standard field theories, we model the horizon as a fractal space. We postulate a fundamental polarization-stacking factor $\alpha = 4$, which defines the Hausdorff dimension D_H:

$$D_H = 3\frac{\ln(2)}{\ln(\alpha)} = 3\frac{\ln(2)}{\ln(4)} = \frac{3}{2} \tag{5}$$

Crucially, to maintain accordance with the Bekenstein-Hawking bounds and the holographic principle, the informational entropy S scales proportionally with the boundary area A, rather than internal volume. The modified action for the framework utilizes a Hausdorff measure to properly weight the discrete manifold:

$$S_{\text{fractal}} = \int dt \int d^{3/2}x \left(\frac{1}{2}\ell_p^{1/2} h_{(1)}^{\mu\nu} \Box_{(n)}^{3/4} h_{\mu\nu}^{(1)} + h_{(1)}^{\mu\nu} R_{\mu\nu}^{(2)} - h_{(1)}^{\mu\nu} \mathcal{K}_{\mu\nu}^{(n)} \right) \tag{6}$$

By forcing the momentum space integration to obey this $D_H = 3/2$ restriction, the loop integral for vacuum energy becomes effectively bottlenecked:

$$\int_0^\Lambda k^{3/2-1} dk = \int_0^\Lambda k^{1/2} dk \propto \Lambda^{3/2} \tag{7}$$

This demonstrates that the fractal packing strictly limits high-energy modes to $\Lambda^{3/2}$, thereby curing the ultraviolet divergence without violating holographic area-entropy laws.

4 Discrete Time Accumulation and Thermodynamic Bridging

Time (t) is strictly defined as an emergent property: the discrete accumulation of entangled polarization layers. Each frozen layer adds precisely two units of von Neumann entropy ($I(n) = 2\ln 2$).

This discrete entropy is linked to the continuous curvature via a thermodynamic bridge ($\delta Q = T\delta S$). The energy flux crossing the horizon is bound to the integral of the quadratic polarization tensor $R_{\mu\nu}^{(2)}$ along the null horizon generator k^μ:

$$T_n(2\ln 2) = \int \lambda R_{\mu\nu}^{(2)} k^\mu k^\nu d\lambda d^2 A \tag{8}$$

The emergent time coordinate at layer N is the sum of this mutual information:

$$t(N) = \tau_0 \sum_{n=1}^{N} 2\ln(2) \tag{9}$$

Where τ_0 is the fundamental temporal unit. The irreversibility of this sum formally defines the Arrow of Time. To mathematically bridge this temporal unit with the $D_H = 3/2$ fractal geometry, we define the spatiotemporal lattice spacing δx^λ. Each step corresponds to a discrete layer:

$$\delta x^\lambda = f(\tau_0, 2\ln 2)u^\lambda \tag{10}$$

5 Topological Matter and Quantized Mass

Matter is proposed to be a coherent column of polarization layers (a knot) formed across the stacked entanglement. To preserve the Bianchi identities across a discrete fractal space, we utilize a Fractional Finite Difference Operator ($\Delta_\lambda^{3/4}$).

The Topological Knot Tensor $\mathcal{K}_{\mu\nu}^{(N)}$ is derived by taking fractional steps of the trace-free quadratic polarization across the lattice gaps, directed by the background flow vector field u^λ:

$$\mathcal{K}_{\mu\nu}^{(N)} = u^\lambda \frac{\Delta^{3/4}}{\delta x^\lambda} \left(R_{\mu\nu}^{(2)} - \frac{1}{4} g_{\mu\nu} R^{(2)} \right) \tag{11}$$

This transforms the interior coherent equation into a balance of rank-2 tensors operating within the discrete geometry:

$$\ell_p^{1/2} \Box_{(n)}^{3/4} h_{\mu\nu}^{(1,n)} + R_{\mu\nu}^{(2,n)} = \mathcal{K}_{\mu\nu}^{(n)} \tag{12}$$

Integrating over the N layers, the rest mass M accumulates entropically:

$$M(N) = m_0 \sum_{n=1}^{N} S_n = m_0 N(2\ln 2) \tag{13}$$

Furthermore, the particle spin J is modeled as the conserved topological current (winding number w_n). Because non-local wave propagation enforces a closed contour phase shift, fractional Stokes' theorem dictates that the winding number is strictly quantized:

$$J = \frac{\hbar}{2\pi} \sum_{n=1}^{N} w_n \tag{14}$$

6 Simulation Results and Methodology

[Note: This section is reserved for the ongoing numerical testing of the Osacra-Gravaser Framework.]

Due to the extreme nonlinear coupling of the tensor networks dictated by the binding matrix, analytic solutions break down across extended manifolds. To simulate the peeling layers and the discrete time evolution of the ρ_{horizon} density matrix, we employ distributed tensor-network algorithms over a localized cloud architecture.

This environment allows for the aggressive containerization of MERA (Multi-scale Entanglement Renormalization Ansatz) networks. By isolating each discrete energy shell n, we mitigate computational memory overhead while calculating the Winding Number (w_n) and Knot Tensor ($\mathcal{K}_{\mu\nu}^{(n)}$) iteratively layer by layer.

6.1 Current Benchmarks